

Experiment - 03.

- Aim : Perform text cleaning, perform lemmatization (any method), remove stop words, label encoding. Create representations using TF-IDF. Save outputs. Dataset : <https://github.com/PICT-NLP/BE-NLP-Elective/blob/main/3-preprocessing/News-dataset.pickle>.

• Theory :

(1) Introduction

- Text preprocessing is an essential step in Natural Language Processing (NLP). That prepares raw textual data for machine learning models.
- Raw text often contains noise such as punctuation, stop words, inconsistent casing, and irrelevant tokens. Cleaning and transforming text improves model accuracy and computational efficiency.

(2) Dataset Description -

- The dataset is a pickled news dataset containing textual news articles along with their category labels. The goal is to preprocess the textual content and convert it into numerical form suitable for machine learning.

(3). Methodology -3.1. Text Cleaning

- Text cleaning removes unwanted characters and standardizes the text.
- Steps performed.
 - Convert text to lowercase.
 - Remove punctuation.
 - Remove numbers.
 - Remove extra white space.
- Mathematically, text cleaning can be represented as a transformation:

$$T_{\text{clean}} = f(T_{\text{raw}})$$

T_{raw} = original text.
 f = cleaning function.

3.2. Stop words removal

- Stop words are common words such as the, is, and of that do not carry significant meaning.
 Let, $T = \{w_1, w_2, \dots, w_n\}$

- After removing stop words $T' = T - S$.
 S = set of stop words.
 T' = filtered text.

- This reduces dimensionality and improves feature quality.

3.3. Lemmatization -

- Lemmatization converts words to their base or dictionary form.
- Examples:
 - running $\rightarrow$ run
 - better $\rightarrow$ good.
- Formally,

$$L(w) = \text{lemma}(w)$$
- Where,
 - w = word.
 - $L(w)$ = lemma of the word.
- It preserves semantic meaning better than Stemming.

3.4. Label Encoding.

- Machine learning models require numerical labels. Therefore, categorical labels are converted into integers.
- If categories are,

{ Sports, Politics, Business }

- Then encoding may be:
Sports $\rightarrow$ 0, Politics $\rightarrow$ 1, Business $\rightarrow$ 2.

- This transformation is represented as -
 $y_{\text{encoded}} = g(y_{\text{original}})$.

Where g is the encoding function.

3.5. TF-IDF Representation -

TF-IDF converts text into numerical feature vectors based on word importance.

- It combines:

Term Frequency (TF) - Measures how often a word appears in a document.

$$TF(t,d) = \frac{f_{t,d}}{\sum_k f_{k,d}}$$

- Where,

$f_{t,d}$ = frequency of term t in document d .

- Inverse Document Frequency (IDF) - Measures how important a word is across documents.

$$IDF(t) = \log \frac{N}{df_t}$$

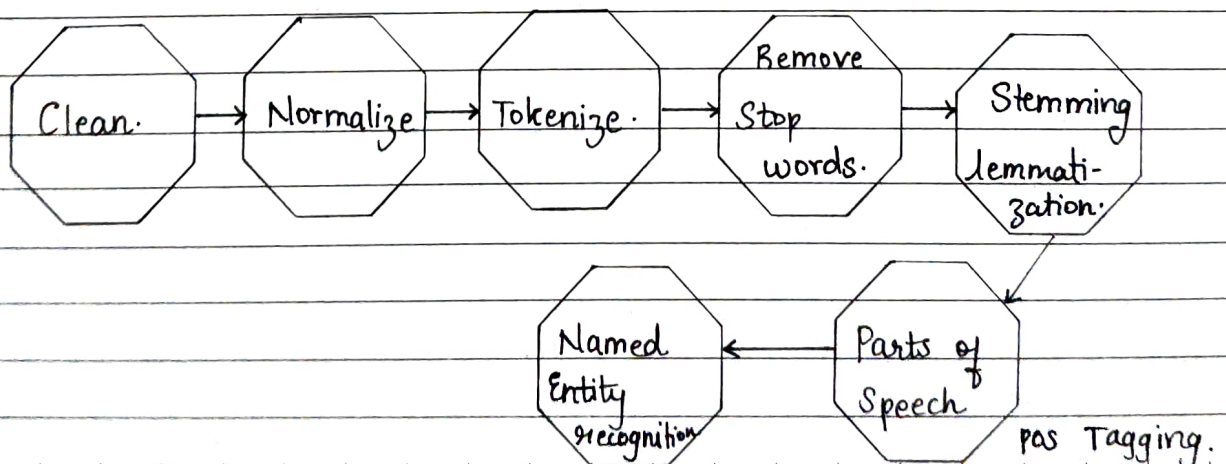
Where, N = total documents.
 df_t = documents containing term t .

- TF-IDF Formula

$$TF-IDF(t,d) = TF(t,d) \times IDF(t)$$

This gives higher weight to rare but meaningful words.

- * Text Preprocessing Steps -



(4) Saving Output -

After preprocessing, the following outputs are stored:

- cleaned text dataset.
- Encoded label array.
- TF-IDF feature matrix.
- Vocabulary list.

Saving outputs improves efficiency and reproducibility.

Common storage formats include:

- CSV for cleaned text.
- Pickle for TF-IDF vectorizer and matrices.
- Numpy arrays for labels.

(5) Advantages -

- Reduces noise in data.
- Improves classification accuracy.
- Standardizes textual input.
- Converts categorical labels to machine readable formats.
- Highlights important terms using TF-IDF.
- Reduces computational complexity.

(6) Applications -

- News classification.
- Sentiment analysis.

- Fake news detection.
- Topic modelling.
- Recommendation systems.
- Cyberbullying detection

(7). Limitations -

- TFIDF ignores word order and context.
- High dimensionality for large vocabularies.
- Lemmatization can be computationally expensive.
- Stop word removal may sometimes remove meaningful words.

• CONCLUSION:

The processed dataset becomes suitable for machine learning models such as Naive Bayes, Logistic regression, Support vector Machines, and deep learning approaches leading to improved performance and reliable predictions. Thus, we successfully studied and implemented text cleaning.

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